

F100 / WATER FILTER

Seven components. One pressure-boundary weld.

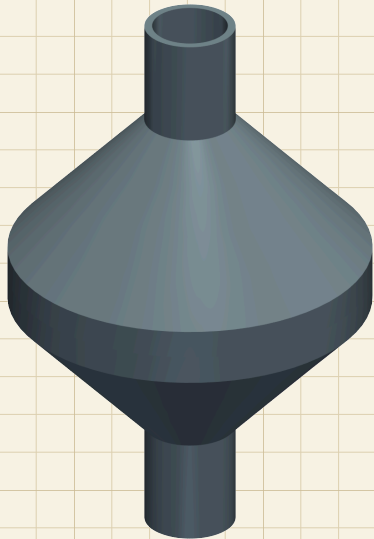
DESIGN INTENT

[insert text here]

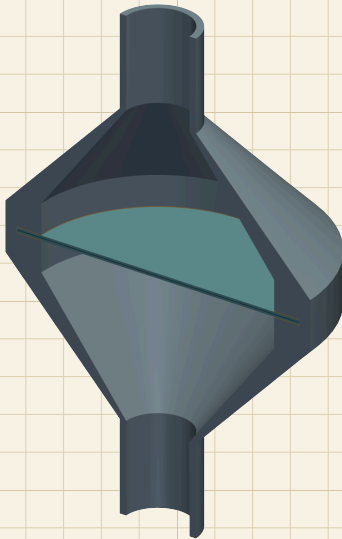
PROJECT	Water filtration
DOCUMENT	F100-SIMPLE-001
AUTHOR	Engineering concept
CHECKED	Review required
REV	S0
DATE	2026-09-14

F100 / ASSEMBLED ISOMETRIC AND CUTAWAY

ASSEMBLED ISOMETRIC



HOUSING AND PACK / CUTAWAY ISO



CALCULATION CONSTANTS

Edit the three Excel files in input/, then run uv run doc.py. Constants drive CAD and calculations; requirements reference those constants. Update water properties consistently when changing temperature.

CONSTANTS

Symbol	Value / unit	Definition / basis
temperature _F	70 degF	Service water temperature; User
ρ_w	997.99 kg/m3	Water density at stated temperature; IAPWS approximation; pressure effects omitted
μ_w	0.000979 Pa s	Water dynamic viscosity; IAPWS approximation
$\dot{m}_n$	0.14 lb/s	Nominal mass flow; User
$\dot{m}_s$	0.47 lb/s	Surge mass flow; User
p_{meop}	650 psi	Housing operating pressure difference; User: internal minus external
p_{proof}	1300 psi	Housing proof pressure difference; User
p_{burst}	2600 psi	Housing burst pressure difference; User
dp _{clean_limit}	5 psi	Clean pressure-drop limit; User
dp _{loaded_limit}	18 psi	Loaded pressure-drop limit; User; nominal flow basis
dirt _{mass}	0.318 g	Retained dry dirt mass; User; contaminant unspecified
mass _{limit}	1 lb	Maximum dry hardware mass; User
rating	5 um	Filtration rating target; User; test definition pending
efficiency	99.9 percent	Capture efficiency target; User
efficiency _{size}	3 um	Efficiency particle size; User
active _d	58 mm	Frame opening / exposed screen diameter; Design
frame _d	70 mm	Frame and mesh outside diameter; Design
frame _{raw_t}	0.2 mm	Uncoined sheet thickness; Design
frame _{assembled_t}	0.17 mm	Nominal coined frame envelope; Unvalidated manufacturing assumption
coarse _t	0.48 mm	Each coarse envelope thickness; Calender target; raw catalog 0.508 mm
fine _t	0.18 mm	Dutch-twill envelope thickness; Placeholder; NOT supplier-confirmed Dutch twill
body _d	76 mm	Housing outside diameter; Simplified design
pocket _d	70.4 mm	Single frame seat outer diameter; Design
outer _{shoulder_z}	6 mm	Outer cone start from weld plane; Design
inner _{shoulder_z}	8 mm	Inner cone start from weld plane; Design
stub _{start_z}	36 mm	Cone end / straight stub start; Design
half _{length}	58 mm	Housing half length; Design
stub _{od}	19.05 mm	Integral weld stub outside diameter; Design; compatible titanium lines
stub _{id}	15.75 mm	Flow bore diameter; Design
C_{vf}	6e+07 1/m	Fine Dutch-twill viscous resistance; Assumed; fit finished cloth data
C_{if}	3000 1	Fine Dutch-twill inertial resistance; Assumed; not transferred supplier data
C_{vc}	200000 1/m	Combined coarse viscous resistance; Assumed
C_{ic}	40 1	Combined coarse inertial resistance; Assumed
K_{body}	3 1	Combined housing loss coefficient; Assumed; changed housing not flow-validated
α_c	1e+09 m/kg	Specific cake resistance; Sensitivity input; contaminant uncharacterized
ρ_{ti}	4510 kg/m3	CP titanium Grade 2 density; TIMET
ρ_{ss}	8000 kg/m3	Stainless frame density; Engineering nominal
Sy	275 MPa	Ti Grade 2 minimum yield screening value; TIMET / ATI; stock certificate required
Su	345 MPa	Ti Grade 2 minimum tensile screening value; ATI
coarse _{areal_mass}	1.367 kg/m2	Each coarse cloth areal mass; TWP 0.280 lb/ft2 rounded
fine _{areal_mass}	0.95 kg/m2	Dutch-twill areal mass; Placeholder; NOT supplier-confirmed Dutch twill
growth _{factor}	1.2 1	Dry mass development reserve multiplier; Design assumption

01 / REQUIREMENTS AND INTERPRETATION

[insert text here]

REQUIREMENT-BY-REQUIREMENT ASSESSMENT

ID / requirement	Target	Assessment
R01 / Housing MEOP	650 psi	[insert text here]
R02 / Housing proof	1300 psi	[insert text here]
R03 / Housing burst	2600 psi	[insert text here]
R04 / Micron rating	5 um	[insert text here]
R05 / Efficiency	99.9 percent at 3 um	[insert text here]
R06 / Nominal flow	0.14 lb/s	[insert text here]
R07 / Surge flow	0.47 lb/s	[insert text here]
R08 / Clean pressure drop	5 psi	[insert text here]
R09 / Retained dirt	0.318 g	[insert text here]
R10 / Loaded pressure drop	18 psi	[insert text here]
R11 / Dry mass	1 lb	[insert text here]
R12 / Water temperature	70 degF	[insert text here]
R13 / Component count	Seven	[insert text here]

02 / ARCHITECTURE

[insert text here]

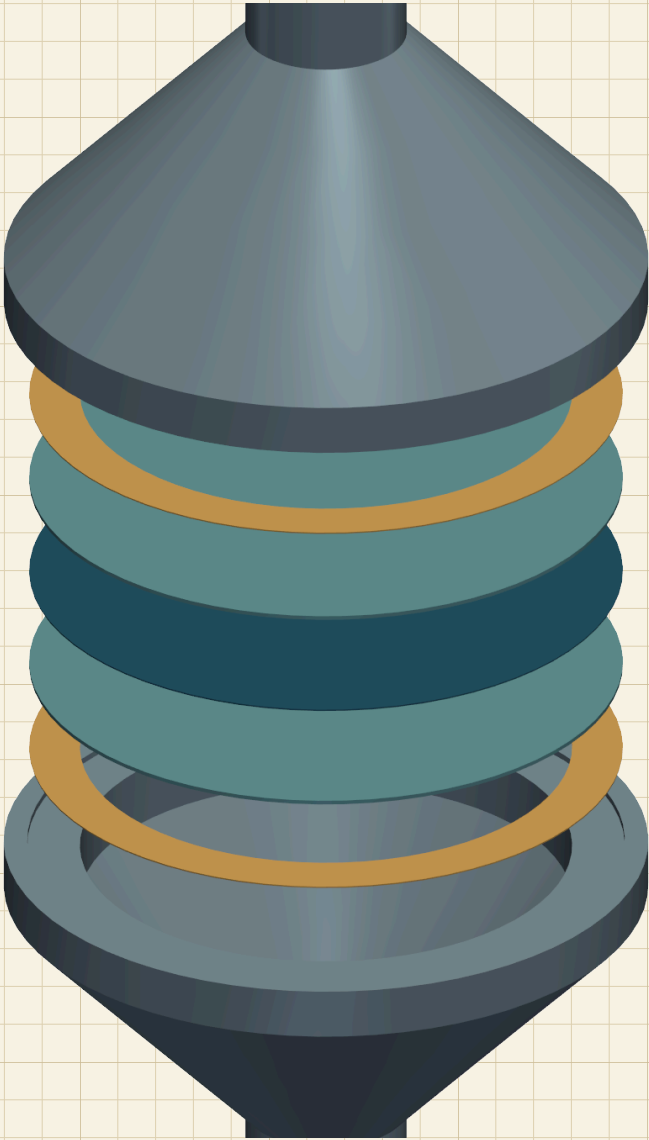
SEVEN-PIECE BILL OF MATERIALS

Component	Qty	Definition
Housing half	2	CP Ti Grade 2; identical; integral weld stubs
Annular frame	2	316L; OD 70 mm, ID 58 mm, raw thickness 0.2 mm
Coarse support	2	40 x 40; assembled thickness 0.48 mm
Fine cloth	1	Dutch twill candidate; thickness 0.18 mm assumed

03 / ASSEMBLY EXPLODED

[insert text here]

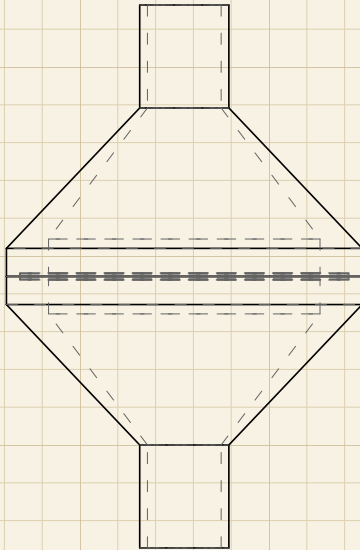
SEVEN COMPONENTS



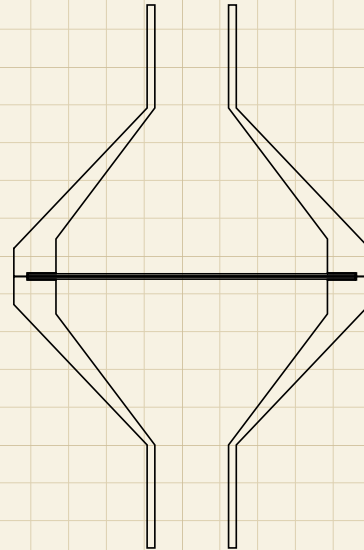
[insert text here]

ALL FOUR DRAWINGS

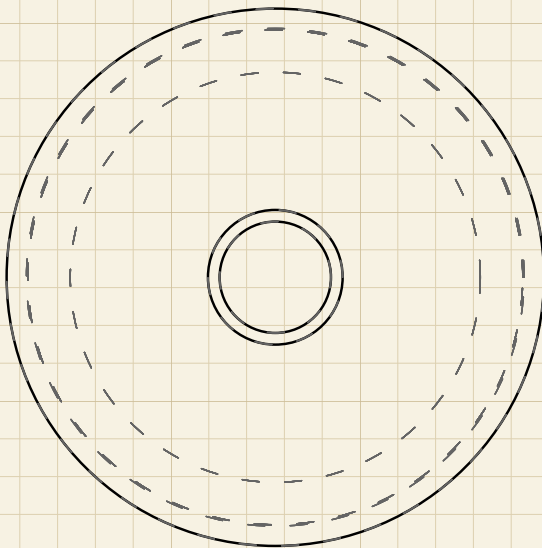
FRONT



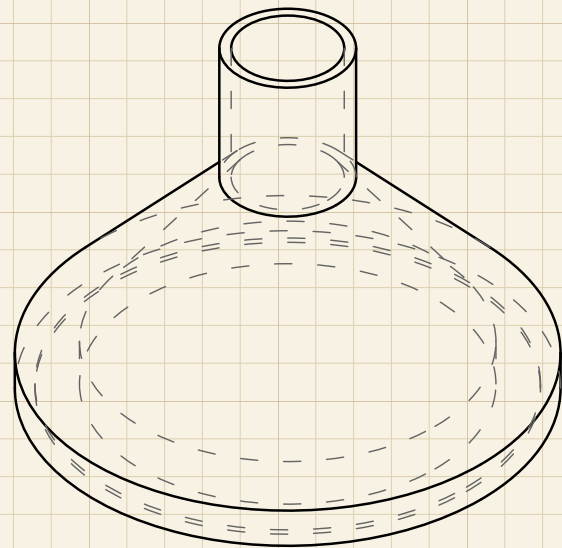
SECTION A-A / CENTER PLANE



END



ONE HOUSING HALF / USE TWICE



06 / FILTER ELEMENT

[insert text here]

MESH SOURCING

Supplier / item		Catalog basis	Assessment
GKD / Dutch-twilled family		5 um end of family; 316L listed; exact construction needs RFQ	[insert text here]
TWP / 040X040T0100		Coarse 40 x 40, 0.010 in wire; T316	[insert text here]
Darby / 40316.010PL		Coarse 40 x 40, 0.010 in wire; T316	[insert text here]
Dorstener / custom fabrication		Custom cloth and element fabrication lead	[insert text here]

SOURCE TRAIL

[insert text here]

[\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#)

08 / FLOW BASIS AND CLEAN-SCREEN EQUATION

[insert text here]

WATER AND GEOMETRY

Symbol	Value
ρ_w	997.990 kg/m ³
μ_w	0.000979 Pa s
d_{face}	58.000 mm
d_{bore}	15.75 mm
$\dot{m}_n$	0.140 lb/s
$\dot{m}_s$	0.470 lb/s

EXPPOSED AREA AND APPROACH VELOCITY

$$A_{\text{face}} = \pi \cdot \frac{(d_{\text{face}})^2}{4} = \pi \cdot \frac{(58.000 \text{ mm})^2}{4} = 2642.079 \text{ mm}^2 \tag{1}$$

$$U_n = \frac{\dot{m}_n}{\rho_w \cdot A_{\text{face}}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.024 \frac{\text{m}}{\text{s}} \tag{2}$$

$$U_s = \frac{\dot{m}_s}{\rho_w \cdot A_{\text{face}}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.081 \frac{\text{m}}{\text{s}} \tag{3}$$

$$U_s = 0.081 \text{ m/s}$$

SCREEN-SPECIFIC RESISTANCE MODEL

$$\text{dp}_{\text{screen}} = C_i \cdot \rho \cdot U^2 + C_v \cdot U \cdot \mu \tag{4}$$

USE OF THE SUPPLIED EQUATIONS

[insert text here]

$$C_v = A_1 \frac{L}{D_i^2}$$
$$C_i = A_2 \frac{L}{D_a}$$

09 / SIMPLIFIED SURGE ESTIMATION

[insert text here]

ASSUMED RESISTANCE COEFFICIENTS

Use the assumed values below for Dutch twill and the combined coarse pair. These are preliminary inputs, not measured supplier data. The viscous terms scale with viscosity and velocity; the inertial terms scale with density and velocity squared. Housing loss uses bore velocity:

$$\Delta p_{\text{body}} = K_{\text{body}} \rho \frac{V_{\text{bore}}^2}{2}.$$

COEFFICIENT DEFINITIONS / CONSTANTS WORKBOOK

Symbol	Assumed value	Definition
C_{vf}	6e+07 1/m	Viscous resistance of the single fine Dutch-twill layer
C_{if}	3000 1	Dimensionless inertial resistance of the fine layer
C_{vc}	200000 1/m	Combined viscous resistance of both coarse supports
C_{ic}	40 1	Combined inertial resistance of both coarse supports
K_{body}	3 1	Dimensionless inlet, outlet and transition loss; excludes mesh

SURGE ASSEMBLY PRESSURE DROP

$$A_{\text{bore}} = \pi \cdot \frac{(d_{\text{bore}})^2}{4} = \pi \cdot \frac{(15.750 \text{ mm})^2}{4} = 194.828 \text{ mm}^2 \quad (5)$$

$$V_{\text{bore}} = \frac{\dot{m}}{\rho_w \cdot A_{\text{bore}}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 194.828 \text{ mm}^2} = 1.096 \frac{\text{m}}{\text{s}} \quad (6)$$

$$U = \frac{\dot{m}}{\rho_w \cdot A_{\text{face}}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.081 \frac{\text{m}}{\text{s}} \quad (7)$$

$$\text{DP} = (C_{vf} + C_{vc}) \cdot \mu_w \cdot U + (C_{if} + C_{ic}) \cdot \rho_w \cdot (U)^2 + K_{\text{body}} \cdot \rho_w \cdot \frac{(V_{\text{bore}})^2}{2} \quad (8)$$

$$= (60000000.000 \frac{1}{\text{m}} + 200000.000 \frac{1}{\text{m}}) \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot 0.081 \frac{\text{m}}{\text{s}} + (3000.000 + 40.000) \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot (0.081 \frac{\text{m}}{\text{s}})^2 + 3.000 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(1.096 \frac{\text{m}}{\text{s}})^2}{2} \quad (9)$$

$$= 3.829 \text{ psi} \quad (10)$$

DP = 3.829 psi

NOMINAL ASSEMBLY PRESSURE DROP

$$A_{\text{bore}} = \pi \cdot \frac{(d_{\text{bore}})^2}{4} = \pi \cdot \frac{(15.750 \text{ mm})^2}{4} = 194.828 \text{ mm}^2 \quad (11)$$

$$V_{\text{bore}} = \frac{\dot{m}}{\rho_w \cdot A_{\text{bore}}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 194.828 \text{ mm}^2} = 0.327 \frac{\text{m}}{\text{s}} \quad (12)$$

$$U = \frac{\dot{m}}{\rho_w \cdot A_{\text{face}}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.024 \frac{\text{m}}{\text{s}} \quad (13)$$

$$\text{DP} = (C_{vf} + C_{vc}) \cdot \mu_w \cdot U + (C_{if} + C_{ic}) \cdot \rho_w \cdot (U)^2 + K_{\text{body}} \cdot \rho_w \cdot \frac{(V_{\text{bore}})^2}{2} \quad (14)$$

$$= (60000000.000 \frac{1}{\text{m}} + 200000.000 \frac{1}{\text{m}}) \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot 0.024 \frac{\text{m}}{\text{s}} + (3000.000 + 40.000) \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot (0.024 \frac{\text{m}}{\text{s}})^2 + 3.000 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(0.327 \frac{\text{m}}{\text{s}})^2}{2} \quad (15)$$

$$= 0.484 \text{ psi} \quad (16)$$

DP = 0.484 psi

11 / LOADED PRESSURE DROP

[insert text here]

SERIES-CAKE EQUATION

[insert text here]

$$\Delta p_l = C_v \mu U + C_i \rho U^2 + \mu U \alpha_c \frac{m_d}{A_f}$$

CAKE INPUTS / CONSTANTS TABLE

Symbol	Value
α_c	1e+09 m/kg
m_d	0.318 g

LOADED ASSEMBLY PRESSURE DROP

$DP_{loaded_n} = DP_n + \mu_w \cdot U_n \cdot \alpha_c \cdot \frac{m_d}{A_{face}} = 0.4842499437890715 \text{pound}_{force\,per\,square\,inch} + 0.001 \text{ Pa} \cdot \text{s} \cdot 0.024 \frac{\text{m}}{\text{s}} \cdot 1000000000.000 \frac{\text{m}}{\text{kg}} \cdot \frac{0.318 \text{ g}}{2642.079 \text{ mm}^2} = 0.896 \text{ psi} \quad (17)$

$DP_{loaded_s} = DP_s + \mu_w \cdot U_s \cdot \alpha_c \cdot \frac{m_d}{A_{face}} = 3.82863240161482 \text{pound}_{force\,per\,square\,inch} + 0.001 \text{ Pa} \cdot \text{s} \cdot 0.081 \frac{\text{m}}{\text{s}} \cdot 1000000000.000 \frac{\text{m}}{\text{kg}} \cdot \frac{0.318 \text{ g}}{2642.079 \text{ mm}^2} = 5.210 \text{ psi} \quad (18)$

DP_{loaded,s} = 5.210 psi

12 / HOUSING PRESSURE SCREENING

[insert text here]

PRESSURE AND WELD-RIM SECTION

Input	Value
MEOP	650 psi
Proof	1300 psi
Burst	2600 psi
r_i	35.200 mm
r_o	38.000 mm
S_y	275 MPa
S_u	345 MPa

CLOSED-END THICK-CYLINDER CHECK AT THE WELD RIM

$$\sigma_h = p_{burst} \cdot \frac{(r_o)^2 + (r_i)^2}{(r_o)^2 - (r_i)^2} \tag{19}$$

$$= 17.93 \text{ MPa} \cdot \frac{(38.00 \text{ mm})^2 + (35.20 \text{ mm})^2}{(38.00 \text{ mm})^2 - (35.20 \text{ mm})^2} \tag{20}$$

$$= 234.67 \text{ MPa} \tag{21}$$

$$\sigma_z = p_{burst} \cdot \frac{(r_i)^2}{(r_o)^2 - (r_i)^2} = 17.93 \text{ MPa} \cdot \frac{(35.20 \text{ mm})^2}{(38.00 \text{ mm})^2 - (35.20 \text{ mm})^2} = 108.37 \text{ MPa} \tag{22}$$

$$\sigma_r = -17.93 \text{ MPa} \tag{23}$$

$$\sigma_{vm} = \left(\frac{(\sigma_h - \sigma_z)^2 + (\sigma_z - \sigma_r)^2 + (\sigma_r - \sigma_h)^2}{2} \right)^{0.5} \tag{24}$$

$$= \left(\frac{(234.67 \text{ MPa} - 108.37 \text{ MPa})^2 + (108.37 \text{ MPa} - -17.93 \text{ MPa})^2 + (-17.93 \text{ MPa} - 234.67 \text{ MPa})^2}{2} \right)^{0.5} \tag{25}$$

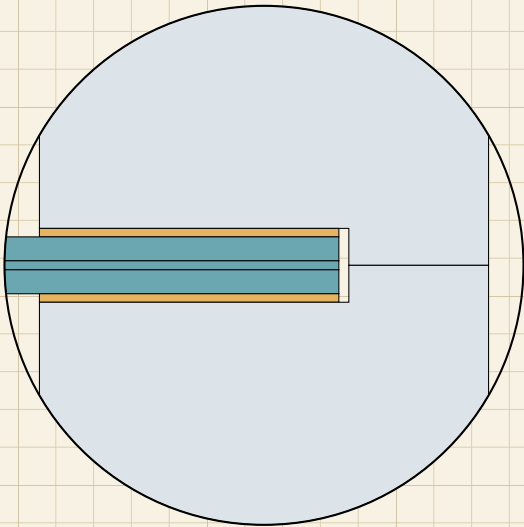
$$= 218.75 \text{ MPa} \tag{26}$$

$\sigma_{vm} = 218.75 \text{ MPa}$

DETAIL B

[insert text here]

HOUSING AND PACK / SECTION DETAIL



Circular section from Build123d. Gold: solid frames; blue: media envelopes.

22 / FULL MASS CALCULATIONS

[insert text here]

DENSITY, SHEET AND CLOTH INPUTS / CONSTANTS TABLE

Input	Value
V_{half}	33552.778 mm ³

DRY FINISHED MASS WITH A DEVELOPMENT RESERVE

$$A_{pack} = \pi \cdot \frac{(d_{pack})^2}{4} = \pi \cdot \frac{(70.000 \text{ mm})^2}{4} = 3848.451 \text{ mm}^2 \tag{27}$$

$$V_{frame} = \pi \cdot \left((d_{pack})^2 - (d_{face})^2 \right) \cdot \frac{t_{frame}}{4} = \pi \cdot ((70.000 \text{ mm})^2 - (58.000 \text{ mm})^2) \cdot \frac{0.200 \text{ mm}}{4} = 241.274 \text{ mm}^3 \tag{28}$$

$$m_{housings} = 2 \cdot V_{half} \cdot \rho_{ti} = 2 \cdot 33552.778 \text{ mm}^3 \cdot 4510.000 \frac{\text{kg}}{\text{m}^3} = 302.646 \text{ g} \tag{29}$$

$$m_{frames} = 2 \cdot V_{frame} \cdot \rho_{ss} = 2 \cdot 241.274 \text{ mm}^3 \cdot 8000.000 \frac{\text{kg}}{\text{m}^3} = 3.860 \text{ g} \tag{30}$$

$$m_{coarse} = 2 \cdot A_{pack} \cdot w_{coarse} = 2 \cdot 3848.451 \text{ mm}^2 \cdot 1.367 \frac{\text{kg}}{\text{m}^2} = 10.522 \text{ g} \tag{31}$$

$$m_{fine} = A_{pack} \cdot w_{fine} = 3848.451 \text{ mm}^2 \cdot 0.950 \frac{\text{kg}}{\text{m}^2} = 3.656 \text{ g} \tag{32}$$

$$m_{dry} = m_{housings} + m_{frames} + m_{coarse} + m_{fine} = 302.646 \text{ g} + 3.860 \text{ g} + 10.522 \text{ g} + 3.656 \text{ g} = 0.707 \text{ lb} \tag{33}$$

$$m_{reserve} = \text{growth} \cdot m_{dry} = 1.200 \cdot 0.707 \text{ lb} = 0.848 \text{ lb} \tag{34}$$

$$\text{MS}_{mass} = \frac{m_{limit}}{m_{reserve}} - 1 = \frac{1.000 \text{ lb}}{0.848 \text{ lb}} - 1 = 0.179 \tag{35}$$

MS_{mass} = 0.179

MASS SCOPE

[insert text here]

24 / REFERENCES

[insert text here]

REFERENCES

- [1] RPD HIFLO-S / 5-S catalog row, Haver & Boecker — Accessed 2026-09-14. 5-S: 850 x 380, 5 μ m geometric pore, zeta 9020, 0.18 mm, 0.95 kg/m², 34% porosity. No beta-3 certification or applicable water flow curve supplied. oh.haverboecker.com/woven-wire-filters/weave-ty...
- [2] GKD / Dutch-twilled family — 5 μ m end of family; 316L listed; exact construction needs RFQ www.gkd-group.com/wp-content/uploads/2024/09/br...
- [3] TWP / 040X040T0100 — Coarse 40 x 40, 0.010 in wire; T316 www.twpinc.com/40-mesh-t316-stainless-010-wire...
- [4] Darby / 40316.010PL — Coarse 40 x 40, 0.010 in wire; T316 catalog.darbywiremesh.com/item/filtration-and-s...
- [5] Dorstener / custom fabrication — Custom cloth and element fabrication lead dwt-inc.com
- [6] TIMETAL 50A / CP titanium Grade 2, TIMET — Density 4.51 g/cm³; guaranteed 275 MPa minimum yield; typical modulus range 105–120 GPa. Do not use typical tensile strength as a minimum. www.timet.com/documents/datasheets/cp-grades/ti...
- [7] ATI CP Grade 2 / UNS R50400, ATI — Listed minimum tensile 345 MPa and yield 275 MPa. Actual housing stock size, product form and material certificate require confirmation. www.atimaterials.com/Products/Pages/ati-cp2.aspx
- [8] Supra range / 316L ASTM A240 properties, Outokumpu, Table 6 — 170 MPa minimum yield and 485 MPa minimum tensile for listed 316L sheet/plate forms; not a woven-wire allowable. www.outokumpu.com/-/media/files/products/supra...
- [9] Orbital Welding System / Series 40, Swagelok — Head family accommodates the 82 mm diameter range. Diameter compatibility does not prove axial clearance or qualify this titanium weld. www.swagelok.com/downloads/webcatalogs/en/ms-02...
- [10] Orbital Welding User's Manual, Swagelok — Process and shielding terminology; no prequalified F100 weld schedule is claimed. www.swagelok.com/downloads/WebCatalogs/en/MS-13...
- [11] J. C. Armour and J. N. Cannon, Fluid Flow Through Woven Screens, AIChE Journal, 1968, 14(3), 415–420 — DOI 10.1002/aic.690140315. Formula conventions used here are those provided by the user. aiche.onlinelibrary.wiley.com/doi/abs/10.1002/a...
- [12] Supplementary Release on Properties of Liquid Water, IAPWS, 2011 — Ambient-pressure property approximation at 70 F; final qualification should account for pressure dependence. www.iapws.org/relguide/LiquidWater.pdf
- [13] ASTM F316 pore characterization, ASTM International — Membrane scope; a woven-metal method needs applicability review. Pore characterization alone does not establish particle retention. store.astm.org/standards/f316
- [14] ISO 16889:2022 / multi-pass filter evaluation, ISO — Method reference, not a claim that the proposed water/3 μ m challenge complies with this hydraulic-fluid standard. www.iso.org/standard/77245.html
- [15] Garrett / project input, User-supplied design brief and pressure-drop equations, 2026 — Governing architecture and equations supplied in task. Clarification: housing gauge pressure, not blocked-element differential; water at 70 F. Original NASA report identifier/file not supplied, so its table entries are treated as user-transcribed data.